A Corrected Integrodifference Competition Model in a Shifting Habitat

Xiang Huang Liang Kong

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Abstract

We formulate a positivity-preserving integrodifference competition model in a habitat moving at constant speed. Signed environmental variation is placed in the low-density growth factor, while density dependence remains nonnegative. We establish the local order structure, invariance of the unit-square state space, preservation of the competitive product order, the exact action of dispersal on exponential modes, an exponential comparison principle, and the geometry of the favorable moving corridor. We then give three global results with explicit hypotheses: uniform extinction in a sufficiently fast habitat, a certified intermediate-speed exclusion theorem, and a certified weak-competition coexistence theorem. The exclusion result uses a finite linear block spreading certificate. The coexistence result derives eventual positive floors from one such certificate per species and then proves convergence in every strictly smaller certified corridor. Identifying the certified fronts with sharp variational spreading speeds remains a separate conditional step. For the exclusion theorem, the published scalar inside-spreading theorem of Li–Bewick–Barnard–Fagan supplies the mathematical route, but its full joined exponential–sine lower solution has not yet been formalized here. The exact convolution, phase-shift, and compact-truncation identities for each flank are formalized. An appendix audits the original proposal and records, by category, which claims fail as stated and how they can be repaired.

1 Introduction

Integrodifference equations are natural models for populations with discrete generations and nonlocal dispersal. If favorable habitat moves at speed $c > 0$, persistence requires the population to disperse and grow fast enough to remain in the favorable region. For two competing species, the same problem also depends on invasion at the two semi-trivial equilibria and on the order structure of the nonlinear update.

The purpose of this note is twofold. First, we give a corrected model for which nonnegative initial data generate nonnegative solutions and the two-species update is genuinely competitive. Second, we prove global extinction, exclusion, and coexistence statements while separating explicit lower-spreading certificates from the subsequent interior squeezing argument. The distinction between certified fronts and sharp fronts remains important: empty-habitat spreading speeds control upper fronts, but they do not by themselves determine coexistence or competitive exclusion.

2 Model and assumptions

For $i = 1, 2$, let $k_i : \mathbb{R} \to [0, \infty)$ be a continuous integrable probability density,

$$\int_{\mathbb{R}} k_i(z)\, dz = 1.$$

Let $\rho_i\colon \mathbb{R} \to (-1, \infty)$ be continuous and nondecreasing, with finite limits

$$-1 < \rho_i^- := \lim_{s\to-\infty} \rho_i(s) < 0 < \rho_i^+ := \lim_{s\to+\infty} \rho_i(s).$$

The species may have distinct environmental responses; the specialization $\rho_1 = \rho_2$ is allowed. Let $\alpha_1, \alpha_2 \geq 0$.

For $\rho > -1$ and $\alpha, u, v \geq 0$, define

$$\widehat{F}(\rho, \alpha; u, v) := \frac{(1+\rho)u}{1 + [\rho]_+(u + \alpha v)}, \qquad [\rho]_+ := \max\{\rho, 0\}. \tag{1}$$

The corrected two-species system is

$$u_{n+1}(x) = \int_{\mathbb{R}} k_1(x-y)\widehat{F}(\rho_1(y-cn), \alpha_1; u_n(y), v_n(y))\, dy, \tag{2}$$

$$v_{n+1}(x) = \int_{\mathbb{R}} k_2(x-y)\widehat{F}(\rho_2(y-cn), \alpha_2; v_n(y), u_n(y))\, dy. \tag{3}$$

In favorable habitat, where $\rho \geq 0$, this is the usual Beverton–Holt competition response

$$\widehat{F}(\rho, \alpha; u, v) = \frac{(1+\rho)u}{1 + \rho(u + \alpha v)}.$$

In unfavorable habitat, where $-1 < \rho \leq 0$, it becomes

$$\widehat{F}(\rho, \alpha; u, v) = (1+\rho)u.$$

Thus unfavorable habitat gives subcritical linear growth instead of negative density dependence.

3 Local dynamics and competitive order

Proposition 3.1 (Positivity, bounds, and local order)**.** *Suppose $\rho > -1$ and $\alpha, u, v \geq 0$. Then*

(i) $\widehat{F}(\rho, \alpha; u, v) \geq 0$*;*

(ii) $\widehat{F}(\rho, \alpha; u, v) \leq (1+\rho)u$*;*

(iii) if $0 \leq u \leq 1$*, then* $0 \leq \widehat{F}(\rho, \alpha; u, v) \leq 1$*;*

(iv) $\widehat{F}$ *is nondecreasing in* u *and nonincreasing in* v*.*

Proof. Put $b = [\rho]_+$ and $D = 1 + b(u + \alpha v)$. Since $b, u, \alpha, v \geq 0$, one has $D \geq 1$. The numerator is nonnegative because $1 + \rho > 0$, which proves (i). Dividing the numerator by $D \geq 1$ proves (ii).

For (iii), suppose first that $\rho \leq 0$. Then $b = 0$, and hence

$$\widehat{F}(\rho, \alpha; u, v) = (1+\rho)u \leq u \leq 1.$$

If $\rho \geq 0$, then

$$D - (1+\rho)u = (1-u) + \rho\alpha v \geq 0,$$

which again gives $\widehat{F} \leq 1$.

Finally, on the positive cone,

$$\frac{\partial \widehat{F}}{\partial u} = \frac{(1+\rho)(1 + b\alpha v)}{D^2} \geq 0, \qquad \frac{\partial \widehat{F}}{\partial v} = -\frac{(1+\rho)ub\alpha}{D^2} \leq 0.$$

This proves (iv). □

Proposition 3.2 (Favorable coexistence equilibrium)**.** *Suppose* $0 < \alpha_1, \alpha_2 < 1$*, and define*

$$u^* = \frac{1-\alpha_1}{1-\alpha_1\alpha_2}, \qquad v^* = \frac{1-\alpha_2}{1-\alpha_1\alpha_2}.$$

Then $u^*, v^* > 0$*,*

$$u^* + \alpha_1 v^* = 1, \qquad v^* + \alpha_2 u^* = 1,$$

and for every $\rho \geq 0$*,*

$$\widehat{F}(\rho, \alpha_1; u^*, v^*) = u^*, \qquad \widehat{F}(\rho, \alpha_2; v^*, u^*) = v^*.$$

Proof. Since $0 < \alpha_1\alpha_2 < 1$, all displayed denominators are positive. Direct substitution gives the two nullcline identities. If $u + \alpha v = 1$ and $\rho \geq 0$, then

$$\widehat{F}(\rho, \alpha; u, v) = \frac{(1+\rho)u}{1+\rho} = u.$$

Applying this identity to the two nullclines proves the result. □

The low-density multiplier of a rare species in favorable habitat, when its competitor is at density one, is

$$m(\rho, \alpha) := \frac{1+\rho}{1+\rho\alpha}, \qquad \rho > 0. \tag{4}$$

Proposition 3.3 (Weak and strong competition at a resident equilibrium)**.** *Let* $\rho > 0$*.*

(i) If $0 \leq \alpha < 1$*, then* $m(\rho, \alpha) > 1$*.*

(ii) If $\alpha > 1$*, then* $m(\rho, \alpha) < 1$*.*

Proof. The denominator in (4) is positive. The first inequality is equivalent to $\rho\alpha < \rho$, and the second is equivalent to $\rho < \rho\alpha$. □

For bounded continuous states, write

$$(u_1, v_1) \preceq_{\mathrm{c}} (u_2, v_2) \quad \Longleftrightarrow \quad u_1(x) \leq u_2(x) \text{ and } v_1(x) \geq v_2(x) \quad \text{for every } x \in \mathbb{R}.$$

This is the competitive product order. Let $\mathcal{T}_n$ denote one update of (2)–(3).

Theorem 3.4 (State-space invariance and competitive comparison)**.** *Assume the kernels and environments satisfy the hypotheses of Section 2. Then* $\mathcal{T}_n$ *maps bounded continuous* $[0,1]^2$*-valued states to bounded continuous* $[0,1]^2$*-valued states. Moreover,*

$$(u_1, v_1) \preceq_{\mathrm{c}} (u_2, v_2) \quad \Longrightarrow \quad \mathcal{T}_n(u_1, v_1) \preceq_{\mathrm{c}} \mathcal{T}_n(u_2, v_2).$$

Proof. Proposition 3.1 bounds each source term between zero and one. Integration against a nonnegative kernel of mass one preserves these bounds. Bounded continuity of the new profiles follows from the standard continuity theorem for convolution of an L^1 kernel with a bounded continuous function.

For the order statement, increasing the focal profile and decreasing the competitor profile increases the corresponding local source by Proposition 3.1(iv). Multiplication by a nonnegative kernel and integration preserve the inequality. Applying this once to the u-equation and once, with the two order directions reversed, to the v-equation proves the claim. □

Corollary 3.5 (An absent species remains absent)**.** *If* $v_0 \equiv 0$*, then* $v_n \equiv 0$ *for every* $n \in \mathbb{N}$*. The analogous statement holds for* u*.*

Proof. Equation (1) gives $\widehat{F}(\rho, \alpha; 0, u) = 0$. Hence $v_n \equiv 0$ implies $v_{n+1} \equiv 0$, and induction completes the proof. □

4 Dispersal, exponential modes, and upper barriers

For a kernel K, define

$$(\mathcal{D}_K f)(x) := \int_{\mathbb{R}} K(x-y) f(y)\, dy, \qquad \mathcal{M}_K(\mu) := \int_{\mathbb{R}} K(z) e^{\mu z}\, dz.$$

Lemma 4.1 (Exponential eigenfunction identity)**.** *Whenever $\mathcal{M}_K(\mu)$ is finite,*

$$\mathcal{D}_K(e^{-\mu \cdot})(x) = \mathcal{M}_K(\mu) e^{-\mu x}.$$

Proof. With $z = x - y$,

$$\begin{aligned} \int_{\mathbb{R}} K(x-y) e^{-\mu y}\, dy &= \int_{\mathbb{R}} K(z) e^{-\mu(x-z)}\, dz \\ &= e^{-\mu x} \int_{\mathbb{R}} K(z) e^{\mu z}\, dz. \end{aligned}$$

□

Proposition 4.2 (Exact exponential orbit)**.** *Suppose $\mathcal{M}_K(\mu)$ is finite. Let $Lf = R\mathcal{D}_K f$, and let*

$$W_n(x) = A\big(R\mathcal{M}_K(\mu)\big)^n e^{-\mu x}.$$

Then $W_{n+1} = LW_n$. Along $x = sn$,

$$W_n(sn) = A\left(R\mathcal{M}_K(\mu) e^{-\mu s}\right)^n.$$

If $R\mathcal{M}_K(\mu) > 0$, $\mu > 0$, and

$$s > \frac{\log(R\mathcal{M}_K(\mu))}{\mu},$$

then $W_n(sn) \to 0$.

Proof. The recurrence follows from Lemma 4.1. The moving-frame formula is an algebraic rearrangement. The speed inequality is equivalent to

$$0 < R\mathcal{M}_K(\mu) e^{-\mu s} < 1,$$

so the geometric factor tends to zero. □

Theorem 4.3 (Exponential comparison)**.** *Assume $K \geq 0$ and $R > 0$. Fix $\mu > 0$ with $0 < \mathcal{M}_K(\mu) < \infty$, and suppose a nonnegative orbit w_n satisfies*

$$w_{n+1}(x) \leq R\mathcal{D}_K w_n(x).$$

Suppose $A \geq 0$ and

$$w_0(x) \leq A e^{-\mu x} \qquad (x \in \mathbb{R}).$$

Then

$$w_n(x) \leq A\big(R\mathcal{M}_K(\mu)\big)^n e^{-\mu x} \qquad (n \in \mathbb{N},\ x \in \mathbb{R}).$$

Consequently, if

$$s > \frac{\log(R\mathcal{M}_K(\mu))}{\mu},$$

then

$$\sup_{x \geq sn} w_n(x) \longrightarrow 0.$$

Proof. The dispersal operator preserves pointwise order because $K \geq 0$. Induct on n. The initial inequality is the base case. If the bound holds at time n, then

$$w_{n+1} \leq R\mathcal{D}_K w_n \leq R\mathcal{D}_K W_n = W_{n+1}.$$

For $x \geq sn$, the exponential is decreasing, so

$$0 \leq w_n(x) \leq A\left(R\mathcal{M}_K(\mu)e^{-\mu s}\right)^n.$$

The right-hand side is a geometric sequence with ratio strictly between zero and one. ☐

Corollary 4.4 (Componentwise scalar domination)**.** *Suppose $\rho_i(s) \leq \rho_i^+$ for all s, and set $R_i = 1 + \rho_i^+$. Each component of (2)–(3) satisfies*

$$u_{n+1} \leq R_1\mathcal{D}_{k_1}u_n, \qquad v_{n+1} \leq R_2\mathcal{D}_{k_2}v_n.$$

Thus Theorem 4.3 applies separately to both species.

Proof. By Proposition 3.1,

$$\widehat{F}(\rho_i(s), \alpha; u, v) \leq (1 + \rho_i(s))u \leq (1 + \rho_i^+)u.$$

Multiplication by a nonnegative kernel and integration preserve the inequality. ☐

Remark 4.5. Every bounded, compactly supported nonnegative initial profile lies below $Ae^{-\mu x}$ for a suitable A. Hence the preceding upper-front estimate applies to the initial data normally used in spreading theory. It is an upper-spreading result; persistence below the critical speed requires a separate lower-solution argument.

5 The favorable moving corridor

Lemma 5.1 (Moving-coordinate geometry)**.** *Let the habitat move at speed c.*

(i) If an observer moves at speed $s < c$, then $(s - c)n \to -\infty$.

(ii) If $s > c$, then $(s - c)n \to +\infty$.

(iii) If $x_n \geq (c + \varepsilon)n$ for some $\varepsilon > 0$, then $x_n - cn \to +\infty$.

Proof. The first two statements follow from the sign of $s - c$. For the third,

$$x_n - cn \geq \varepsilon n \longrightarrow +\infty.$$

☐

Corollary 5.2 (Uniformly favorable corridor geometry)**.** *Let $\rho(s) \to \rho^+$ as $s \to +\infty$. If*

$$(c + \varepsilon)n \leq x_n \leq (c_{\text{front}} - \varepsilon)n$$

for every n, where $\varepsilon > 0$, then

$$\rho(x_n - cn) \longrightarrow \rho^+.$$

The corridor is nonempty whenever $c + 2\varepsilon \leq c_{\text{front}}$.

Proof. The first conclusion follows by composing the far-right limit of ρ with Lemma 5.1(iii). The nonemptiness condition is the inequality

$$c + \varepsilon \leq c_{\text{front}} - \varepsilon.$$

□

Thus convergence to a far-right favorable equilibrium belongs in a one-sided corridor

$$(c + \varepsilon)n \leq x \leq (c_{\text{front}} - \varepsilon)n.$$

A symmetric band $|x - cn| \leq \eta n$ contains $x = (c - \eta)n$, whose moving coordinate tends to $-\infty$.

6 Certified global results

For later use, set

$$R_i := 1 + \rho_i^+, \qquad \mathcal{M}_i(\mu) := \int_{\mathbb{R}} k_i(z) e^{\mu z}\, dz,$$

and let

$$\mathcal{A}_i := \{\mu > 0 : \mathcal{M}_i(\mu) < \infty\}.$$

When $\mathcal{A}_i \neq \varnothing$, define the admissible linear speed

$$c_i^{\text{lin}} := \inf_{\mu \in \mathcal{A}_i} \frac{\log(R_i \mathcal{M}_i(\mu))}{\mu}. \tag{5}$$

The restriction to $\mathcal{A}_i$ is essential: no logarithm or infimum is silently taken over an infinite exponential moment.

For the opposite direction, set

$$\mathcal{A}_{i,-} := \{\mu > 0 : \mathcal{M}_i(-\mu) < \infty\},$$

and, when this set is nonempty, define

$$c_{i,-}^{\text{lin}} := \inf_{\mu \in \mathcal{A}_{i,-}} \frac{\log(R_i \mathcal{M}_i(-\mu))}{\mu}. \tag{6}$$

The corresponding left boundary in physical coordinates has velocity $-c_{i,-}^{\text{lin}}$.

For $f > c$ and $\varepsilon > 0$, write

$$\mathcal{C}_n(c, f, \varepsilon) := [(c + \varepsilon)n, (f - \varepsilon)n]. \tag{7}$$

For real numbers L, R, a, b, θ, define the expanding interval

$$\mathcal{I}_k(L, R, a, b, \theta) := [L + (a + \theta)k,\ R + (b - \theta)k]. \tag{8}$$

Theorem 6.1 (Fast-habitat extinction; corrected Theorem 2.1)**.** *Assume the hypotheses of Section 2, $c \geq 0$, and $\mathcal{A}_i \neq \varnothing$ for $i = 1, 2$. Let the initial state be continuous and $[0,1]^2$-valued. The following componentwise statements hold independently:*

$$\left.\begin{matrix} u_0(x) = 0 \quad (x > L_1), \\ c > c_1^{\text{lin}} \end{matrix}\right\} \implies \|u_n\|_\infty \longrightarrow 0,$$

and

$$\left.\begin{aligned} v_0(x) = 0 \quad (x > L_2),\\ c > c_2^{\mathrm{lin}} \end{aligned}\right\} \quad \Longrightarrow \quad \|v_n\|_\infty \longrightarrow 0.$$

In particular, if both right-support conditions hold and

$$c > \max\{c_1^{\mathrm{lin}}, c_2^{\mathrm{lin}}\},$$

then

$$\sup_{x\in\mathbb{R}} \max\{u_n(x), v_n(x)\} \longrightarrow 0.$$

Compact support of each initial component is a sufficient, but stronger, version of the two right-support assumptions above.

Proof. We prove the assertion for one component w_n, with kernel k_i. Choose

$$c_i^{\mathrm{lin}} < s < c.$$

By the definition of the infimum, there is $\mu \in \mathcal{A}_i$ such that

$$\frac{\log(R_i \mathcal{M}_i(\mu))}{\mu} < s.$$

The right-support assumption gives $w_0(x) \le Ae^{-\mu x}$ for a finite A. Theorem 4.3 therefore yields constants $B > 0$ and $0 < \lambda < 1$ such that

$$\sup_{x \ge sn} w_n(x) \le B\lambda^n.$$

Choose q with $1 + \rho_i^- < q < 1$. Since $s < c$ and ρ_i is nondecreasing, for all sufficiently large n,

$$y \le sn \quad \Longrightarrow \quad 1 + \rho_i(y - cn) \le q.$$

Split the source integral at $y = sn$. On the left part, the corrected response is at most $q\, w_n(y)$; on the right part, it is at most $R_i w_n(y)$. Positivity and unit mass of the kernel give

$$\|w_{n+1}\|_\infty \le q\|w_n\|_\infty + R_i B\lambda^n.$$

Iteration of this scalar recurrence shows $\|w_n\|_\infty \to 0$. The competitor never enters the upper estimate, so the argument applies separately to both components. $\square$

Definition 6.2 (Finite linear block certificate)**.** For $\sigma \ge 0$, set

$$(\mathcal{L}_\sigma \phi)(x) := \int_{\mathbb{R}} k_2(x-y)\, \sigma\phi(y)\, dy.$$

Fix an integer $m \ge 1$, numbers $z, \eta > 0$, a minimum width W, and displacements $a \le b$. These data form a finite linear block certificate if $z \le \eta$ and, for every interval $[A, B]$ with $B - A \ge W$, the iterates

$$\phi_{A,B}^0 = z\,\mathbf{1}_{[A,B]}, \qquad \phi_{A,B}^{t+1} = \mathcal{L}_\sigma \phi_{A,B}^t$$

are integrable where needed, satisfy

$$0 \le \phi_{A,B}^t \le \eta \qquad (0 \le t \le m),$$

and obey the block expansion estimate

$$\phi_{A,B}^m(x) \ge z \qquad (A + a \le x \le B + b). \tag{9}$$

Proposition 6.3 (Reference-interval reduction)**.** *Assume $k_2 \geq 0$, $\sigma \geq 0$, $W \geq 0$, and $a \leq b$. Once the nonnegativity, upper-bound, and integrability conditions in Definition 6.2 have been checked, it is enough to verify the single estimate*

$$\phi^m_{0,W}(x) \geq z \qquad (a \leq x \leq W + b).$$

If k_2 is integrable, nonnegative, and has mass one, nonnegativity and all the required integrability conditions are automatic, while the universal upper bound follows from the finite scalar checks

$$\sigma^t z \leq \eta \qquad (0 \leq t \leq m).$$

If also $\sigma \geq 1$, it is enough to check the terminal scalar inequality. Moreover, linear homogeneity gives

$$\phi^m_{0,W} = z\, \psi^m_{0,W},$$

where $\psi^0_{0,W} = \mathbf{1}_{[0,W]}$. Hence the reference expansion condition is exactly the dimensionless estimate

$$\psi^m_{0,W}(x) \geq 1 \qquad (a \leq x \leq W + b).$$

Proof. Given $B - A \geq W$ and $x \in [A + a, B + b]$, one can choose

$$H \in [A, B - W] \quad \text{with} \quad x \in [H + a, H + W + b].$$

For example, take

$$H = \max\{A, x - (W + b)\}.$$

Translation invariance sends the reference estimate to the subinterval $[H, H + W]$. Its indicator is bounded above by the indicator of $[A, B]$; positivity of the kernel propagates this order through all m linear steps. Hence (9) holds on the full interval. Convolution preserves integrability, and multiplication by the bounded finite orbit preserves integrability of every comparison integrand. Finally, unit kernel mass gives $\phi^t_{A,B} \leq \sigma^t z$ by induction. Monotonicity of powers handles all earlier times when $\sigma \geq 1$, and linearity gives the displayed scaling identity. □

Proposition 6.4 (Exact total mass)**.** *Suppose k_2 is integrable and has mass one. If $A \leq B$, then every linear interval orbit in Definition 6.2 satisfies*

$$\int_{\mathbb{R}} \phi^t_{A,B}(x)\, dx = \sigma^t z(B - A) \qquad (t \geq 0).$$

Proof. The initial interval floor has integral $z(B - A)$. For integrable functions, Fubini's theorem and the unit-mass identity give

$$\int_{\mathbb{R}} \mathcal{L}_\sigma \phi = \sigma \left(\int_{\mathbb{R}} k_2 \right) \left(\int_{\mathbb{R}} \phi \right) = \sigma \int_{\mathbb{R}} \phi.$$

Induction on t proves the formula. □

Remark 6.5. When $\sigma > 1$, Proposition 6.4 certifies exact exponential growth of the total linearized mass. It does not by itself imply the pointwise reference estimate in Proposition 6.3: the same mass may be distributed too thinly or in the wrong spatial region. The remaining analytic step is therefore a local lower estimate for the relevant convolution power (or, equivalently, a quantitative large-deviation lower estimate), not another integrability or normalization argument.

Theorem 6.6 (Finite-block intermediate-speed exclusion; corrected Theorem 2.2)**.** *Assume the hypotheses of Section 2, $c \ge 0$, and $\|u_n\|_\infty \to 0$. Let the initial state be continuous and $[0,1]^2$-valued, and suppose*

$$k_2(z) = 0 \qquad (|z| > r)$$

for some $r \ge 0$. Choose

$$\rho_0, \delta, z, \eta > 0, \qquad \sigma \ge 0,$$

such that

$$\eta + \alpha_2\delta \le 1, \qquad \sigma \le \frac{1+\rho_0}{1+\rho_0(\eta+\alpha_2\delta)}. \tag{10}$$

Suppose that $(m, z, \eta, W, a, b, \sigma)$ is a finite linear block certificate, and that for some $N \in \mathbb{N}$ and $L, R \in \mathbb{R}$,

$$R - L \ge W, \qquad v_N(y) \ge z \quad (y \in [L, R]). \tag{11}$$

For $k \in \mathbb{N}$, $0 \le t < m$, assume that

$$\phi^t_{L+ak,R+bk}(y) > 0 \quad \Longrightarrow \quad \begin{cases} u_{N+mk+t}(y) \le \delta, \\ \rho_2(y - c(N+mk+t)) \ge \rho_0. \end{cases} \tag{12}$$

Finally, choose

$$0 < \varepsilon_\mathrm{b} < \varepsilon_\mathrm{f} < \varepsilon_\mathrm{t}, \qquad c + 2\varepsilon_\mathrm{t} \le f. \tag{13}$$

and assume the block-speed inequalities

$$a < (c+\varepsilon_\mathrm{b})m, \qquad (f - \varepsilon_\mathrm{b})m < b. \tag{14}$$

Then

$$\|u_n\|_\infty \longrightarrow 0$$

and

$$\sup_{x \in \mathcal{C}_n(c,f,\varepsilon_\mathrm{t})} |v_n(x) - 1| \longrightarrow 0.$$

Proof. On $0 \le q \le \eta$, whenever the two bounds in (12) hold, direct cross multiplication in (1) gives

$$\sigma q \le \widehat{F}(\rho_2, \alpha_2; q, u).$$

Positivity of the kernel and induction on t therefore place each linear iterate $\phi^t_{L+ak,R+bk}$ below the corresponding nonlinear profile. At $t = m$, (9) reproduces the floor z on $[L + a(k+1), R + b(k+1)]$. Induction on k yields a positive floor at all block times. The inequalities (14) place the ε_b-corridor inside these expanding intervals for all large k.

Finite kernel range propagates this floor across the m residue classes of time, losing only a fixed finite amount of corridor width. Thus v_n has an eventual positive floor on the ε_f-corridor at every sufficiently large time. Uniform extinction of u_n, followed by the scalar Beverton–Holt comparison on a still narrower corridor, makes the lower bound converge to one. Since $v_n \le 1$, the required uniform convergence follows. □

Remark 6.7. The front f is certified by the finite convolution estimate (9); it has not yet been identified with a sharp variational spreading speed. The missing analytic step is to derive such a finite block from $f < c_2^{\mathrm{lin}}$, or from another intrinsic linear-determinacy hypothesis. There is a published route for the scalar part: Theorem 2(i)(b) and Lemma 4 of Li–Bewick–Barnard–Fagan [1] construct a positive expanding plateau by joining two truncated exponential–sine flanks. In the notation used here, their general left interior speed is

$$\ell_2 = \max\{c, -c_{2,-}^{\mathrm{lin}}\},$$

not always c. Thus the corridor with left edge $(c+\varepsilon)n$ additionally requires $c \geq -c_{2,-}^{\mathrm{lin}}$; for symmetric kernels and $c \geq 0$, this condition is automatic. Formalizing that scalar lower-solution construction is underway: the exact weighted-moment and compact-truncation identities for each flank, together with its one-step linear subsolution inequality, are proved. Joining the two flanks to a plateau and inserting the small-competitor comparison remain. Completing those steps would close the proposal-level speed implication for corrected Theorem 2.2.

Remark 6.8 (Why the older one-step certificate is too narrow)**.** The valid one-step special case used a kernel minorization

$$k_2(s) \geq \kappa \quad (s \in [a,b]), \tag{15}$$

$$2\theta \leq b - a, \tag{16}$$

$$z \leq \kappa\theta\, \widehat{F}(\rho_0, \alpha_2; z, \delta). \tag{17}$$

Unit kernel mass gives $\kappa\theta \leq 1/2$, so this forces

$$1 \leq \rho_0\big(1 - 2z - 2\alpha_2\delta\big). \tag{18}$$

In particular, $\rho_0 > 1$. Allowing the floor to attenuate during transport and recover only afterward does not solve the geometry: with a kernel supported in $[-r, r]$, zero recovery steps contradict the required growth, while one or more recovery steps erode the right endpoint by at least the entire rightward range. The resulting speed inequality is incompatible with a nonempty positive-width corridor. The simultaneous finite-block recursion avoids both defects by retaining every dispersal path during the block.

Theorem 6.9 (Certified weak-competition coexistence; corrected Theorem 2.3)**.** *Assume the hypotheses of Section 2,*

$$0 < \alpha_1 < 1, \qquad 0 < \alpha_2 < 1, \qquad c \geq 0.$$

Let the initial state be continuous and $[0,1]^2$-valued. Suppose that there are $r \geq 0$ and $S_1, S_2 \in \mathbb{R}$ such that

$$k_i(z) = 0 \quad (|z| > r), \qquad \rho_i(s) = \rho_i^+ \quad (s \geq S_i), \qquad i = 1, 2.$$

Let $0 < \varepsilon_{\mathrm{w}} < \varepsilon_{\mathrm{t}}$ and

$$c + 2\varepsilon_{\mathrm{t}} \leq f.$$

Assume that there are $p_1, p_2 > 0$ and $N_0 \in \mathbb{N}$ such that, for every $n \geq N_0$ and every $x \in \mathcal{C}_n(c, f, \varepsilon_{\mathrm{w}})$,

$$u_n(x) \geq p_1, \qquad v_n(x) \geq p_2. \tag{19}$$

Then

$$\sup_{x \in \mathcal{C}_n(c,f,\varepsilon_{\mathrm{t}})} |u_n(x) - u^*| \longrightarrow 0$$

and

$$\sup_{x \in \mathcal{C}_n(c,f,\varepsilon_{\mathrm{t}})} |v_n(x) - v^*| \longrightarrow 0.$$

In particular, every constant-speed observer $x = sn$ with

$$c + \varepsilon_{\mathrm{t}} \leq s \leq f - \varepsilon_{\mathrm{t}}$$

converges to the favorable coexistence equilibrium.

Proof. Fix a finite depth m. The common finite range of the kernels and the strict margin inequality imply that every predecessor involved in m updates of a target-corridor point remains in the wider corridor for all sufficiently large n. The exact favorable-tail hypothesis makes both environmental responses homogeneous on this predecessor set.

Start the homogeneous competitive recursion from the rectangle

$$[\ell_1, 1] \times [\ell_2, 1],$$

where $0 < \ell_i \leq p_i$ and $\ell_i + \alpha_i \leq 1$. Competitive monotonicity propagates the lower and upper corners of this rectangle through the m exact favorable updates. Under $0 < \alpha_1, \alpha_2 < 1$, the lower corners increase, the upper corners decrease, and both converge to the unique intersection of

$$u + \alpha_1 v = 1, \qquad v + \alpha_2 u = 1,$$

namely (u^*, v^*). Given $\eta > 0$, choose m so that this homogeneous rectangle lies within η of (u^*, v^*), and then choose n large enough for the m-step spatial comparison. This gives the two uniform limits. The observer statement follows by inclusion in the target corridor. □

Remark 6.10. The positive-floor condition (19) is the global persistence input to Theorem 6.9; it is not derived from the empty-habitat moment formulas in this note. The theorem establishes the subsequent interior squeezing to (u^*, v^*) throughout the full weak competition range.

Definition 6.11 (Finite-block corridor certificate)**.** Fix $i \in \{1, 2\}$, a front f, and a wide margin $\varepsilon_{\mathrm{w}} > 0$. Write $w_{1,n} = u_n$ and $w_{2,n} = v_n$. A finite-block corridor certificate for species i consists of

$$m_i, N_i \in \mathbb{N}, \qquad z_i, \eta_i, \sigma_i, W_i, a_i, b_i, L_i, R_i, \varepsilon_i \in \mathbb{R}$$

with $m_i \geq 1$, $0 < z_i \leq \eta_i$, and

$$\eta_i + \alpha_i \leq 1, \qquad 0 \leq \sigma_i \leq \frac{1 + \rho_i^+}{1 + \rho_i^+(\eta_i + \alpha_i)}. \tag{20}$$

The data $(m_i, z_i, \eta_i, W_i, a_i, b_i, \sigma_i)$ must form a finite linear block certificate in the sense of Definition 6.2, with k_2 replaced by k_i. In addition,

$$R_i - L_i \geq W_i, \qquad w_{i,N_i}(y) \geq z_i \quad (y \in [L_i, R_i]), \tag{21}$$

and every positive intermediate linear iterate must remain in the exact favorable tail:

$$\phi^t_{L_i + a_i k, R_i + b_i k}(y) > 0 \quad \Longrightarrow \quad S_i \leq y - c(N_i + m_i k + t) \qquad (0 \leq t < m_i). \tag{22}$$

Finally,

$$0 < \varepsilon_i < \varepsilon_{\mathrm{w}}, \qquad a_i < (c + \varepsilon_i) m_i, \qquad (f - \varepsilon_i) m_i < b_i. \tag{23}$$

Corollary 6.12 (Finite-block coexistence criterion)**.** *Assume all the kernel, favorable-tail, weak-competition, and corridor hypotheses of Theorem 6.9, except* (19)*. If each species has a finite-block corridor certificate in the sense of Definition 6.11, using a common front and common wide margin, then both conclusions of Theorem 6.9 hold.*

Proof. For each species, (20) places the finite linear recursion below the nonlinear response while the competitor is bounded by one. The exact-tail condition and (22) supply the favorable growth bound on the entire positive support of every intermediate iterate. Induction over blocks, followed by finite-range propagation across time residues, produces an eventual floor $w_{i,n} \geq z_i$ on the common wide corridor. Apply Theorem 6.9. □

Definition 6.13 (One-step expanding-floor certificate)**.** Fix $i \in \{1, 2\}$, a front f, and a wide margin $\varepsilon_{\mathrm{w}} > 0$. Suppose that

$$\rho_i(s) = \rho_i^+ \qquad (s \geq S_i).$$

A one-step expanding-floor certificate for species i consists of

$$N_i \in \mathbb{N}, \qquad z_i, \kappa_i, L_i, R_i, a_i, b_i, \theta_i \in \mathbb{R}$$

such that, with $w_{1,n} = u_n$ and $w_{2,n} = v_n$,

$$z_i > 0, \quad \kappa_i \geq 0, \quad \theta_i \geq 0, \quad k_i(s) \geq \kappa_i \quad (s \in [a_i, b_i]), \tag{24}$$

$$b_i - a_i \leq R_i - L_i, \qquad 2\theta_i \leq b_i - a_i, \tag{25}$$

$$z_i + \alpha_i \leq 1, \qquad z_i \leq \kappa_i \theta_i \widehat{F}(\rho_i^+, \alpha_i; z_i, 1), \tag{26}$$

$$w_{i,N_i}(y) \geq z_i \quad (y \in [L_i, R_i]), \tag{27}$$

$$S_i \leq y - c(N_i + k) \quad (k \in \mathbb{N},\ y \in \mathcal{I}_k(L_i, R_i, a_i, b_i, \theta_i)), \tag{28}$$

$$a_i + \theta_i < c + \varepsilon_{\mathrm{w}}, \qquad f - \varepsilon_{\mathrm{w}} < b_i - \theta_i. \tag{29}$$

These conditions give a directly checkable sufficient mechanism for (19); they are not part of the full weak-competition statement of Theorem 6.9.

Corollary 6.14 (A narrower minorization criterion for Theorem 2.3)**.** *Assume all the kernel, favorable-tail, weak-competition, and corridor hypotheses of Theorem 6.9, except* (19)*. If each species has a one-step expanding-floor certificate in the sense of Definition 6.13, with the same f and ε_{w}, then both conclusions of Theorem 6.9 hold.*

Proof. A direct one-step induction, with the competitor bounded by one, gives

$$w_{i,N_i+k}(y) \geq z_i \qquad \big(y \in \mathcal{I}_k(L_i, R_i, a_i, b_i, \theta_i)\big).$$

The strict inequalities in (29) place the wider corridor inside this expanding interval for all sufficiently large times. Thus (19) holds with $p_i = z_i$, after increasing the common starting time. Apply Theorem 6.9. ☐

Remark 6.15 (Feasibility of the one-step certificate)**.** The preceding corollary is deliberately narrower than Theorem 6.9. Since k_i has mass one, (24) and (25) imply

$$\kappa_i \theta_i \leq \frac{1}{2}.$$

Combining this bound with (26) and dividing by $z_i > 0$ gives the sharper necessary condition

$$1 \leq \rho_i^+ \big(1 - 2z_i - 2\alpha_i\big). \tag{30}$$

Consequently, such a certificate requires

$$\alpha_i < \frac{1}{2} \qquad \text{and} \qquad \rho_i^+ > 1.$$

Even under these two necessary inequalities, (30) remains an additional quantitative condition to be checked. This limitation belongs to this particular one-step minorization device, not to the full weak-competition range $0 < \alpha_i < 1$ covered by the positive-floor theorem.

7 Conditional variational-speed forms

Assume in this section that $\mathcal{A}_1, \mathcal{A}_2 \neq \varnothing$. For weak competition, define the favorable rare-species multipliers

$$\widetilde{R}_i := \frac{R_i}{1 + \rho_i^+ \alpha_i}$$

and the admissible invasion-speed candidates

$$\widetilde{c}_i^{\mathrm{lin}} := \inf_{\mu \in \mathcal{A}_i} \frac{\log(\widetilde{R}_i \mathcal{M}_i(\mu))}{\mu}, \qquad c_{\mathrm{coex}}^{\mathrm{lin}} := \min_i \widetilde{c}_i^{\mathrm{lin}}. \tag{31}$$

Corollary 7.1 (Conditional variational form of Theorem 2.2)**.** *Assume*

$$c_1^{\mathrm{lin}} < c < c_2^{\mathrm{lin}}.$$

Fix

$$0 < \varepsilon < \frac{c_2^{\mathrm{lin}} - c}{2}.$$

If the hypotheses of Theorem 6.6 admit a quantitative certificate with

$$f = c_2^{\mathrm{lin}}, \qquad 0 < \varepsilon_{\mathrm{w}} < \varepsilon_{\mathrm{t}} = \varepsilon,$$

then

$$\|u_n\|_\infty \longrightarrow 0, \qquad \sup_{(c+\varepsilon)n \le x \le (c_2^{\mathrm{lin}} - \varepsilon)n} |v_n(x) - 1| \longrightarrow 0.$$

Proof. Apply Theorem 6.6 with the stated values of f and ε_{t}. $\square$

Corollary 7.2 (Conditional variational form of Theorem 2.3)**.** *Assume the kernel, favorable-tail, weak-competition, and state hypotheses of Theorem 6.9. Suppose*

$$0 < c < c_{\mathrm{coex}}^{\mathrm{lin}}.$$

Fix

$$0 < \varepsilon < \frac{c_{\mathrm{coex}}^{\mathrm{lin}} - c}{2}.$$

If a scalar invasion or spreading theorem supplies constants $p_1, p_2 > 0$ and an eventual wider-corridor bound (19) with

$$f = c_{\mathrm{coex}}^{\mathrm{lin}}, \qquad 0 < \varepsilon_{\mathrm{w}} < \varepsilon_{\mathrm{t}} = \varepsilon,$$

then

$$\sup_{(c+\varepsilon)n \le x \le (c_{\mathrm{coex}}^{\mathrm{lin}} - \varepsilon)n} \max\{|u_n(x) - u^*|,\ |v_n(x) - v^*|\} \longrightarrow 0.$$

Proof. Apply Theorem 6.9 with the stated front and margins. $\square$

Remark 7.3 (The remaining sharp-speed step)**.** The moment formulas (5) and (31) do not, by themselves, produce the lower-spreading certificates used above. A separate scalar spreading or linear-determinacy theorem must derive those certificates from more intrinsic assumptions. Only after that step may c_i^{lin} or $\widetilde{c}_i^{\mathrm{lin}}$ be identified with a sharp nonlinear speed usually denoted by c_i^* or $\widetilde{c}_i^*$. Extending the certified theorems from finite-range kernels and, for coexistence, exactly constant favorable tails to more general kernels and asymptotically constant tails is also left open.

8 A reusable generalization

The same local argument applies to the more general response

$$G_i(s;u,v) := \frac{R_i(s)u}{1 + b_i(s)(u + \alpha_i v)}, \qquad R_i(s) > 0, \quad b_i(s) \geq 0. \tag{32}$$

Positivity and competitive order follow immediately. A sufficient condition for $G_i(s;u,v) \leq 1$ whenever $0 \leq u \leq 1$ is

$$R_i(s) \leq 1 + b_i(s).$$

Indeed,

$$1 + b_i(s)(u + \alpha_i v) - R_i(s)u = 1 - \big(R_i(s) - b_i(s)\big)u + b_i(s)\alpha_i v$$

If $R_i(s) \leq b_i(s)$, the right-hand side is at least one. Otherwise $0 < R_i(s) - b_i(s) \leq 1$, so $\big(R_i(s) - b_i(s)\big)u \leq 1$, and the right-hand side is again nonnegative. The model (1) is the special case

$$R_i(s) = 1 + \rho_i(s), \qquad b_i(s) = [\rho_i(s)]_+.$$

This formulation separates low-density growth, density dependence, and interspecific competition. It also permits different environments for the two species, asymmetric kernels, directional moments in higher dimensions, and habitat displacements that vary with time.

A Audit of the original proposal

This appendix concerns the CMBE 2026 proposal *Spatial Dynamics of an Integro-Difference Lotka–Volterra Model in a Shifting Habitat*. The purpose is diagnostic: for each defect, we give the shortest mathematical reason that the original statement cannot stand and then record a repair.

A.1 The printed response is not positive on the stated state space

The response printed in the proposal is

$$F(\rho, \alpha; u, v) := \frac{(1+\rho)u}{1 + \rho(u + \alpha v)}. \tag{33}$$

The hypotheses of Theorems 2.1–2.3 allow arbitrary bounded nonnegative initial data. They do not impose $u + \alpha v \leq 1$. Under the permitted values

$$\rho = -\frac{1}{2}, \qquad \alpha = 0, \qquad u = 3, \qquad v = 0,$$

one obtains

$$F\left(-\frac{1}{2}, 0; 3, 0\right) = \frac{(1/2)3}{1 - (1/2)3} = -3.$$

Thus the local update can send an admissible nonnegative state to a negative value. The model, as stated, is not a positive population dynamical system.

Repair. Either add and prove an invariant state-space restriction strong enough to keep every denominator positive, or change the response. The correction (1) gives positivity on the entire positive cone and is therefore the cleaner repair.

A.2 The simplex remark does not restore competition

The proposal contains a red editorial sentence suggesting

$$1 - u - \alpha v \geq 0.$$

This is not included as a theorem hypothesis, and no invariance proof is given. More importantly, even if the inequality is imposed, the direction of competition remains wrong in unfavorable habitat. For

$$\rho = -\frac{1}{2}, \qquad \alpha = 1, \qquad u = \frac{1}{2},$$

both $v = 0$ and $v = \frac{1}{2}$ satisfy $u + \alpha v \leq 1$, but

$$F\left(-\frac{1}{2}, 1; \frac{1}{2}, 0\right) = \frac{1}{3} < \frac{1}{2} = F\left(-\frac{1}{2}, 1; \frac{1}{2}, \frac{1}{2}\right).$$

Increasing the competitor increases the focal-species response. In fact, where the denominator is positive,

$$\frac{\partial F}{\partial v} = -\frac{(1+\rho)u\rho\alpha}{(1+\rho(u+\alpha v))^2},$$

which is positive when $-1 < \rho < 0$, $u > 0$, and $\alpha > 0$.

Repair. Density dependence must remain nonnegative. Replacing ρ by $[\rho]_+$ in the denominator yields (1); the unfavorable response then becomes subcritical linear growth.

A.3 Theorem 2.1 lacks a valid invariant state space

Theorem 2.1 assumes only that the initial pair is nonnegative, bounded, and compactly supported. The calculation above uses data allowed by exactly those assumptions. Consequently, the theorem is not formulated on a state space preserved by its own update, and the comparison principles invoked in the proposal are unavailable.

This defect is logically prior to the speed calculation. Even a correct formula for c_i^* cannot repair a nonlinear map that leaves the positive cone.

Repair. Use the corrected response and initial values in the invariant unit square $[0, 1]^2$, as in Theorem 6.1. Its proof combines the upper exponential estimate ahead of an intermediate frame with a uniform contraction behind that frame. The upper barrier alone would prove only the upper-front part.

A.4 Theorem 2.2 has a missing-species counterexample

Theorem 2.2 assumes that the pair (u_0, v_0) has nonempty compact support, but it does not assume $v_0 \not\equiv 0$. Choose

$$u_0 \not\equiv 0, \qquad v_0 \equiv 0.$$

Then the pair satisfies the printed initial-data hypothesis. Since the second response has a factor $v_n(y)$, induction gives

$$v_n \equiv 0 \qquad \text{for all } n.$$

Therefore

$$\liminf_{n\to\infty} \inf_{|x-cn|\leq \eta n} v_n(x) = 0$$

for every $\eta > 0$, contradicting the asserted strict positivity.

Repair. Any persistence conclusion for species two must assume $v_0 \not\equiv 0$, together with the positivity or irreducibility conditions needed to spread nonzero mass. The finite-time positive seed in (11) is a quantitative replacement that directly excludes the missing-species case. Its feasibility must still be checked; in particular, it is subject to (18).

A.5 Theorem 2.2 uses the wrong persistence region

The symmetric region in the original theorem is

$$|x - cn| \leq \eta n.$$

Its left boundary is $x = (c - \eta)n$, so in habitat coordinates

$$x - cn = -\eta n \longrightarrow -\infty.$$

The left edge therefore samples the unfavorable limit ρ^-, while the right edge samples the favorable limit. A uniform lower bound over the entire symmetric band cannot be justified by convergence to the favorable single-species equilibrium.

Repair. Use a one-sided interior corridor

$$(c + \varepsilon)n \leq x \leq (f - \varepsilon)n,$$

where f is a certified rightward front. Every sequence in this corridor has habitat coordinate tending to $+\infty$. Theorem 6.6 gives such a statement from an explicit finite linear block certificate. Replacing f by a proposed sharp speed c_2^* is valid only after the additional implication in Corollary 7.1 has been established. Since the proposal does not assume symmetric kernels and separately defines leftward moments, leftward and rightward speeds cannot be conflated.

A.6 A rightward speed does not determine the left interior edge

There is a second, independent asymmetry defect. Suppose a continuous probability kernel is supported in an interval $[a, b]$ with $a > c$. Starting from compact support, every individual moves at least a units per generation. Hence after n generations the entire population lies to the right of its initial left endpoint plus na. It is therefore zero near $x = cn$. On the other hand, if the favorable linear multiplier is $R > 1$, then

$$\mathcal{M}_K(\mu) \geq e^{\mu a}, \qquad \frac{\log(R\mathcal{M}_K(\mu))}{\mu} > a \quad (\mu > 0),$$

so $c^* \geq a > c$. Thus the printed rightward speed inequality can hold while persistence near the habitat speed fails.

The corresponding obstruction is formalized in Lean; the audit lists the two theorem names. If $k_2(z) = 0$ for $z < a$, a nonvacuous m-step block certificate must satisfy

$$ma \leq A_{\text{left}}.$$

The geometry needed for a corridor moving at speed q also requires $A_{\text{left}} < mq$, and therefore forces $a < q$.

Repair. Use both directional speeds. The scalar inside-spreading theorem in [1] applies. Its general left interior edge is

$$\left(\max\{c, -c^*_{2,-}\} + \varepsilon\right) n$$

and the right edge at

$$(c^*_2 - \varepsilon)n.$$

If one wants the simpler left edge $(c + \varepsilon)n$, one must assume $c \geq -c^*_{2,-}$; symmetry of the kernel together with $c \geq 0$ is a standard sufficient specialization.

A.7 Theorem 2.3 quantifies over observers behind the habitat

Theorem 2.3 asserts convergence to the positive favorable equilibrium for every $0 < c' < c^\dagger$, where $c^\dagger > c$. This interval includes all c' with $0 < c' < c$. For such an observer,

$$c'n - cn = (c' - c)n \longrightarrow -\infty.$$

Hence the observer moves linearly far into the unfavorable side of the habitat. The claimed convergence to the far-right positive coexistence equilibrium uses the wrong quantifier range.

Repair. The observer speed must satisfy

$$c < c' < f,$$

or the conclusion should be stated uniformly in the corresponding one-sided certified corridor. Theorem 6.9 uses exactly this geometry.

A.8 Theorem 2.3 uses only empty-habitat speeds

The threshold

$$c < \min\{c^*_1, c^*_2\}$$

uses growth at the empty state. Coexistence, however, requires each species to invade a state occupied by the other species. At resident density one, the rare-species multiplier is

$$\widetilde{R}_i = \frac{1 + \rho_i^+}{1 + \rho_i^+ \alpha_i},$$

not $1 + \rho_i^+$. For every $0 < \alpha_i < 1$, this multiplier is greater than one but strictly smaller than the empty-state multiplier. The associated invasion front can therefore be slower than the empty-habitat front.

Repair. First prove coexistence behind a certified front from genuine positive interior floors, as in Theorem 6.9. The reduced speeds in (31) give a conditional candidate for that front, but a separate scalar spreading or linear-determinacy theorem must produce the required floors before those variational quantities can be called exact speeds. The one-step minorization criterion is only a narrower sufficient device: (30) shows that it cannot cover the whole weak-competition interval.

A.9 The strong-competition conclusion does not follow from speed ordering

The summary states that under strong competition the faster species excludes the other, even when both persist alone. Suppose, in the favorable homogeneous environment, that

$$\alpha_1 > 1, \qquad \alpha_2 > 1.$$

By Proposition 3.3, each rare-species multiplier at the other resident-only equilibrium is less than one:

$$\frac{1+\rho^+}{1+\rho^+\alpha_i} < 1.$$

Thus both semi-trivial equilibria resist invasion. This is the local signature of bistability. Ordering the two empty-habitat speeds does not select a unique winner independently of initial data.

Repair. A strong-competition theorem must specify invasion eigenvalues, wave selection, and initial-data conditions. It may yield conditional exclusion or a bistable traveling front, but speed ordering alone is insufficient.

A.10 The PINN convergence sentence is not an analytical consequence

The proposal suggests that training loss tending to zero should imply convergence to the analytical traveling wave and consistent recovery of its speed. A small sampled residual does not by itself imply either conclusion. One must specify at least:

(i) the function space and norm in which the residual is controlled;

(ii) compactness or coercivity for the network sequence;

(iii) consistency of quadrature for the nonlocal integral;

(iv) stability and uniqueness of the target wave after fixing translation;

(v) identifiability of the wave speed.

Without these ingredients, the PINN is a numerical proposal rather than a convergence theorem.

Repair. State the neural-network component as a computational method and validate it numerically. A mathematical convergence claim should be deferred until a separate approximation theorem supplies the missing hypotheses.

A.11 The status and notation are internally inconsistent

Section 2 says that the three theorems are results "we propose to establish," whereas the summary says "we prove" and the conclusion says "we established." These descriptions cannot all be correct in the same document. In addition, both species equations use ρ_1. This is legitimate only if a common environmental response is intended; otherwise the second occurrence should be ρ_2.

Repair. Label only the statements established under their displayed hypotheses as theorems. Keep sharp-speed versions explicitly conditional until the missing spreading implication is proved, as in Section 7. State explicitly whether the two species share one environmental response or have species-specific functions.

B Summary of repairs

Original issue	Required repair
Signed density dependence	Use nonnegative density dependence, for example $[\rho]_+$, or prove a genuinely invariant alternative state space.
Bounded nonnegative data	Work in the invariant unit square and require every persisting species to be initially nonzero.
Symmetric persistence band	Use a one-sided corridor strictly ahead of the habitat, with a certified rightward front.
Observer range $0 < c' < c^\dagger$	Replace it by an interval $c < c' < f$ lying inside the certified corridor.
Empty-habitat coexistence threshold	Use positive interior floors for the certified theorem; relating the front to invasion multipliers requires a separate linear-determinacy result.
Strong competition	Add invasion and initial-data hypotheses; allow bistability.
PINN convergence	Treat it as numerical methodology unless a separate approximation theorem is proved.

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